

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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Abstract

The first thermal shield of a 4 K cryostat intercepts most of the heat arriving from the 300 K environment. It can be held near 50 K by circulated helium gas or at 77 K by liquid nitrogen. This paper separates that choice into three questions and answers each with a dedicated model. (1) The passive penalty of the warmer shield is computed with a three-dimensional multi-region conjugate heat-transfer model of a representative cryostat section that resolves view-factor radiation and the conduction of a 65-line coaxial harness; the view-factor matrix is shown to require row normalisation in a low-emissivity enclosure, after which the model reproduces the closed-form loads to 4 %. (2) The two coolants are compared in a channelled cold plate at a matched 9.4 W duty over Reynolds numbers 500–10 000 and helium pressures of 1–18 bar, with National Institute of Standards and Technology (NIST) properties, three-grid verification and energy closure below 0.2 %. (3) The routes are priced in room-temperature power with Carnot multipliers and with hardware efficiencies. The warmer shield costs nothing at the first stage, 8.3 against 8.4 W; its penalty is 106 mW at the 4 K stage, 86 % of it harness conduction, so it is a property of the wiring rather than of the cryogen. In the cold plate nitrogen gives five to six times helium’s heat-transfer coefficient and holds the plate within 4 K of its inlet where helium at 1 bar floats 21 K above; the hottest point of the helium plate reaches 77 K only at the top of the laminar range, and at 18 bar helium’s pumping power is 8.5 times below nitrogen’s at matched Reynolds number. The ideal ledger favours nitrogen

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by 21 % for a bare shield and helium by 9 % with multi-layer insulation; with hardware efficiencies and the liquefaction energy of the nitrogen consumed, the equivalent room-temperature power favours helium by 36–51 % at the nominal shield temperatures and by 13–56 % once each route is charged with the plate temperature its own cold plate reaches, the low end at a common laminar flow where the helium plate floats to 67 K. Nitrogen becomes competitive only if the harness conduction to 4 K is cut by about a factor of three. Operating maps, uncertainty bands and the complete case set are provided.

Keywords: cryostat, thermal shield, conjugate heat transfer, liquid nitrogen, helium gas, pulse-tube cryocooler, OpenFOAM

Nomenclature

Roman symbols

A	area [m ²]
c_p	specific heat [J kg ⁻¹ K ⁻¹]
D_h	hydraulic diameter [m]
f	Darcy friction factor [-]
F_{ij}	view factor from face i to face j [-]
h	heat-transfer coefficient [W m ⁻² K ⁻¹]
J	radiosity [W m ⁻²]
k	thermal conductivity [W m ⁻¹ K ⁻¹]
$\dot{m}$	mass flow rate [kg s ⁻¹]
Mo	Mouromtseff number [-]
N	number of MLI layers [-]
NTU	number of transfer units [-]
Nu	Nusselt number hD_h/k [-]
p	pressure [Pa]
Pr	Prandtl number [-]
$\dot{Q}$	heat load [W]
q	heat flux [W m ⁻²]
Re	Reynolds number $\rho U D_h / \mu$ [-]
$\dot{S}_{\text{gen}}$	entropy generation rate [W K ⁻¹]
T	temperature [K]
T_0	heat-rejection temperature, 300 K
U	mean velocity [m s ⁻¹]
$\dot{W}$	wall power [W]
$\dot{W}_p$	pumping power [W]
y^+	wall coordinate [-]

Greek symbols

β	thermal expansion coefficient [K ⁻¹]
ε	emissivity; HX effectiveness [-]
η_i	stage efficiency, fraction of Carnot [-]
κ_{eff}	effective harness conductivity [W m ⁻¹ K ⁻¹]
μ	dynamic viscosity [Pa s]
ρ	density [kg m ⁻³]
σ	Stefan–Boltzmann constant

Subscripts 0 ambient; 1 first stage; 2 second (4 K) stage; b bulk; c conduction; r radiation; sat saturation; w wall.

Acronyms CFD computational fluid dynamics; CHT conjugate heat transfer; DTT Divertor Tokamak Test facility; GCI grid convergence index; ITER International Thermonuclear Experimental Reactor; LHC Large Hadron Collider; LHD Large Helical Device; MLI multi-layer insulation; NIST National Institute of Standards and Technology; NTU number of transfer units; OVC outer vacuum can; PTFE polytetrafluoroethylene; PTR pulse-tube refrigerator; SIMPLE semi-implicit method for pressure-linked equations; SST shear-stress transport; STL stereolithography (surface file format).

1. Introduction

A cryostat that holds a payload at 4 K places at least one thermal shield between the payload and the 300 K vacuum vessel. The shield intercepts the radiation from the vessel and the conduction along supports and wiring, and it does so at a temperature where refrigeration is one to two orders of magnitude cheaper than at 4 K [1, 2, 3]. Two ways of cooling this first shield are in wide use. Large superconducting systems circulate pressurised helium gas through cooling tubes on the shield: the thermal shield of the International Thermonuclear Experimental Reactor (ITER) runs on helium at 80 K and 1.8 MPa [4], the shield of the Divertor Tokamak Test facility (DTT) on the same loop conditions [5], the shields of the Large Helical Device (LHD) on gaseous helium in parallel paths [6], and the shields of the Large Hadron Collider (LHC) on helium at 50–75 K [7, 8]. Magnet and detector cryostats, and the older generation of helium cryostats, use a liquid-nitrogen bath or loop at 77 K [9, 10]; Li et al. made the practical case for it three decades ago, that a boiling cryogen holds the shield temperature constant and carries a stored reserve, whereas a helium-gas shield warms as soon as its refrigerator stops [11]. Compact laboratory cryostats, including the dry dilution refrigerators that host superconducting quantum processors, use the first stage of a helium pulse-tube refrigerator (PTR) at 40–50 K instead [12, 13, 14]. As these laboratory systems grow toward the multi-metre cryostats now being built for quantum platforms [15, 16], they inherit the shield-cooling question that fusion and accelerator cryostats answered with circulated fluids, and the cryogenic plant becomes a dominant part of the system’s energy cost [17, 18, 19].

The published treatments of this choice fall into four strands that do not meet. First, shield design temperatures are optimised with lumped static budgets and refrigeration-power minimisation: Lagrange-multiplier optimisation of nitrogen, neon and helium-gas intercepts on supports and in insulation [11], multi-stage shields for transmission-line cryostats [20], fusion thermal shields [21, 5], and single-shield helium cryostats brought to zero boil-off by load minimisation [22]. These return a stage total or a shield temperature field, from one-dimensional budgets or, for the feeder shields, finite-element models, but not where the wiring load lands or how it couples to the radiation field. Second, wiring loads are measured and modelled per cable [14, 23, 24], again as conduction integrals that assign the harness a single resistance and leave it uncoupled from the radiation field. Third,

surface-to-surface radiation in computational fluid dynamics (CFD) codes has been validated on enclosure benchmarks and applied to fusion vacuum vessels [25], but not coupled to temperature-dependent harness conduction in a cryostat shield, and the closure of agglomerated view-factor matrices in such enclosures has received little attention. Fourth, cryogenic heat exchangers are optimised by entropy-generation minimisation [26, 27, 28] and liquid-nitrogen cooled components are now modelled with combined analytical and CFD frameworks [29], yet helium gas and liquid nitrogen have not been ranked against each other as forced-convection first-shield coolants on a common duty across flow regimes and pressures, with heat-transfer, hydraulic and effectiveness metrics reported together and the single-phase margin of the nitrogen verified. Finally, the cryocooler and liquefier efficiencies that separate an ideal Carnot ledger from a real one are well documented [30, 8, 31], but the two ledgers have not been reconciled for this choice.

The gap is therefore not a missing number but a missing junction. What is absent is a treatment that (i) resolves shield radiation and harness conduction together in three dimensions on one geometry, (ii) compares the two coolants like for like in the cold plate that geometry implies, and (iii) prices both routes with ideal and with hardware efficiencies, so that the conditions under which each route wins can be stated rather than assumed.

This paper therefore asks three questions about a representative 4 K cryostat section with a 65-line coaxial harness (Fig. 1):

Q1 (passive penalty). What does raising the first shield from 50 K to 77 K cost in heat load at the first stage and at the 4 K stage, and which mechanism carries the cost?

Q2 (coolant performance). If the first-stage duty is removed by a coolant circulated through a channelled cold plate, how do helium gas and liquid nitrogen compare across flow regimes, pressures and bases of comparison, and where is each fluid’s best operating point?

Q3 (work ledger). What do the two routes cost in room-temperature power, with ideal Carnot multipliers and with the efficiencies of real hardware?

Q1 is answered with a three-dimensional multi-region conjugate heat-transfer (CHT) model that resolves view-factor radiation and temperature-dependent harness conduction. Its stage-integrated loads are verified against closed-form budgets; its contribution is the spatial distribution of the load and the

split between harness families. Q2 is answered with a grid-converged cold-plate model run for both fluids over $Re = 500\text{--}10^4$ and helium pressures of 1–18 bar, with properties from the NIST reference equations of state [32]. Q3 combines the stage loads of Q1 with Carnot multipliers and with published cryocooler and liquid-nitrogen plant efficiencies [30, 33, 31]. The dilution unit or any payload below 4 K is outside the scope; the results apply to any 4 K cryostat with a wired first shield.

2. Models and methods

2.1. Governing equations

Both models solve steady conservation equations with OpenFOAM v2412 [34]. In the enclosure model the inter-stage gaps are vacuum, so each gap region carries only the radiative exchange and each solid region carries conduction with a temperature-dependent conductivity,

$$\nabla \cdot [\kappa_{\text{eff}}(T) \nabla T] = 0 \quad \text{in the harness bundles,} \quad (1)$$

coupled at the bundle surfaces to the radiative flux of the enclosing gap. Radiation is solved as a grey-diffuse radiosity system on the agglomerated boundary faces,

$$J_i = \varepsilon_i \sigma T_i^4 + (1 - \varepsilon_i) \sum_j F_{ij} J_j, \quad q_{r,i} = J_i - \sum_j F_{ij} J_j, \quad (2)$$

where F_{ij} are geometric view factors computed by ray tracing between agglomerated faces (`viewFactor` model). For two parallel grey surfaces Eq. (2) reduces to the closed form $q_r = \sigma(T_{\text{hot}}^4 - T_{\text{cold}}^4)/(1/\varepsilon_{\text{hot}} + 1/\varepsilon_{\text{cold}} - 1)$ [35], which is used as the verification anchor.

The cold-plate model solves the steady Reynolds-averaged (or, in the laminar range, the plain) continuity, momentum and energy equations for a single-phase coolant,

$$\nabla \cdot (\rho \mathbf{U}) = 0, \quad (3)$$

$$\nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p + \nabla \cdot [(\mu + \mu_t)(\nabla \mathbf{U} + \nabla \mathbf{U}^T)] + \rho \mathbf{g}, \quad (4)$$

$$\nabla \cdot (\rho \mathbf{U} h) = \nabla \cdot \left[\left(\frac{k}{c_p} + \frac{\mu_t}{Pr_t} \right) \nabla h \right], \quad (5)$$

with $\mu_t = 0$ for laminar runs and μ_t from the $k\text{--}\omega$ shear-stress transport (SST) model [36] for $Re \geq 5000$ ($Pr_t = 0.85$). The sweep is run as pure forced

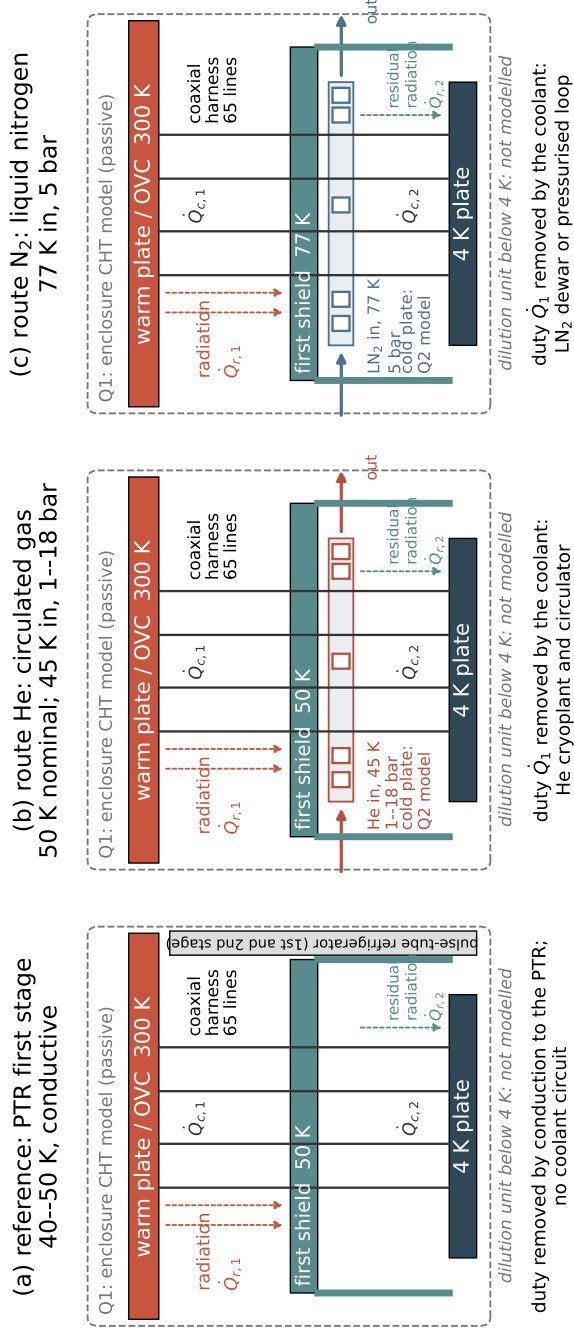


Figure 1: The three first-shield cooling routes and the two models, drawn for the cryostat class studied: a 300 K warm plate and outer vacuum can, a first shield at 50 or 77 K and a 4 K plate, with the dilution unit or payload below 4 K not modelled. (a) Reference architecture with a conductively coupled PTR first stage; (b) circulated helium gas through a channelled cold plate; (c) liquid nitrogen through the same cold plate. The dashed box is the domain of the passive enclosure CHT model (Q1); the shaded channel row is the domain of the active cold-plate model (Q2). The loads $\dot{Q}_1 = \dot{Q}_{r,1} + \dot{Q}_{c,1}$ and $\dot{Q}_2 = \dot{Q}_{r,2} + \dot{Q}_{c,2}$ enter the work ledger ($Q3$), $\dot{W} = \sum_i \dot{Q}_i (T_0 - T_i) / (T_i \eta_i)$, with $\eta_i = 1$ (ideal) or the measured fraction of Carnot (real).

convection ($\mathbf{g} = 0$) so that the ranking does not depend on the orientation of the plate; the buoyancy parameter Gr/Re^2 is reported for every run, and four runs with gravity acting along the heated wall quantify the mixed-convection sensitivity (Section 4). Helium is treated as an ideal gas with variable density, $\rho = pM/(R_u T)$, because its density changes by 5–26 % between inlet and outlet and by up to a factor of 2 between bulk and heated wall; liquid nitrogen is treated with the Boussinesq approximation, $\rho = \rho_0[1 - \beta(T - T_0)]$, for which $\beta\Delta T \leq 0.05$ in every run (largest at $\text{Re} = 500$, where the wall sits 8.7 K above the inlet), within the first-order range of the approximation.

2.2. Enclosure CHT model (Q1)

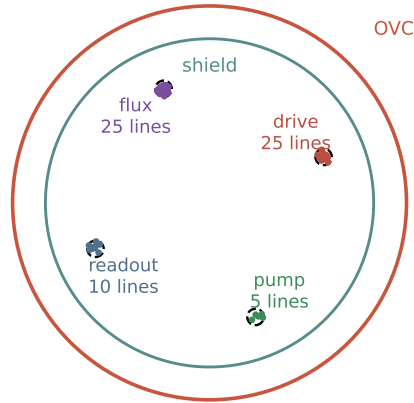
The domain is a $0.718 \times 0.718 \times 0.335$ m section of a cryostat between the warm plate and the first-stage cold plate (Fig. 3; Table 1). Two vacuum gap regions (outer: outer vacuum can (OVC) to shield; inner: warm plate to cold plate and shield) and four solid cylinders of 20 mm diameter spanning the 0.335 m height are meshed with a $30 \times 30 \times 14$ background grid refined to level 2 around the cylinders ($0.2\text{--}0.4 \times 10^6$ cells in total). Each cylinder lumps one functional family of a cabled 65-line control harness [14]: 25 drive, 25 flux, 10 readout and 5 pump lines, all of the UT-085 stainless-steel/PTFE coaxial construction. The family is represented by an effective conductivity

$$\kappa_{\text{eff}}(T) = \frac{n_k}{A_{\text{model}}} \sum_i A_i k_i(T), \quad (6)$$

where n_k is the number of lines in the family, A_i the cross-section of cable component i (outer conductor 1.585 mm^2 and centre conductor 0.205 mm^2 of 304 stainless steel, polytetrafluoroethylene (PTFE) dielectric 2.001 mm^2), $k_i(T)$ the NIST cryogenic conductivity fits, and A_{model} the conduction cross-section of the modelled cylinder in the stage span concerned, calibrated from the mesh so that the resolved conduction reproduces the harness integral $n_k L^{-1} \sum_i A_i \int k_i dT$ (Supplementary Tables S3 and S4 list the areas, lengths, per-bundle targets and the fifth-order polynomial coefficients of κ_{eff} for every bundle and stage span; fit error below 0.5 %; Fig. 2). The stage-integrated conduction is therefore fixed by the cable data by construction, and the model is credited only with its distribution among bundles and its coupling to the radiation field, not with its total.

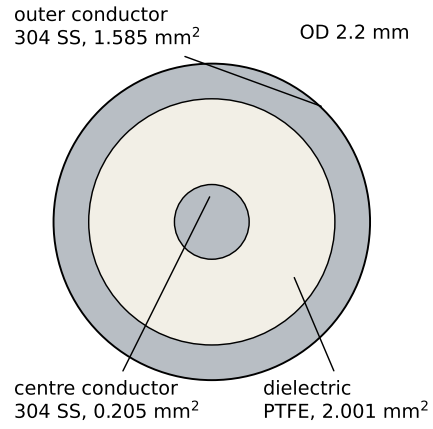
Verification and uncertainty. Stage-integrated radiation is compared with the parallel-plate closed form and harness conduction with the analytic integral; both agree within the tolerances reported in Section 3. Because the

(a) 65 lines in four families



each family → one 20 mm cylinder
at $r = 0.15$ m (plan view, to scale)

(b) UT-085 cable cross-section



$$(c) \kappa_{eff}(T) = \frac{n_k}{A_{model}} \sum_i A_i k_i(T)$$

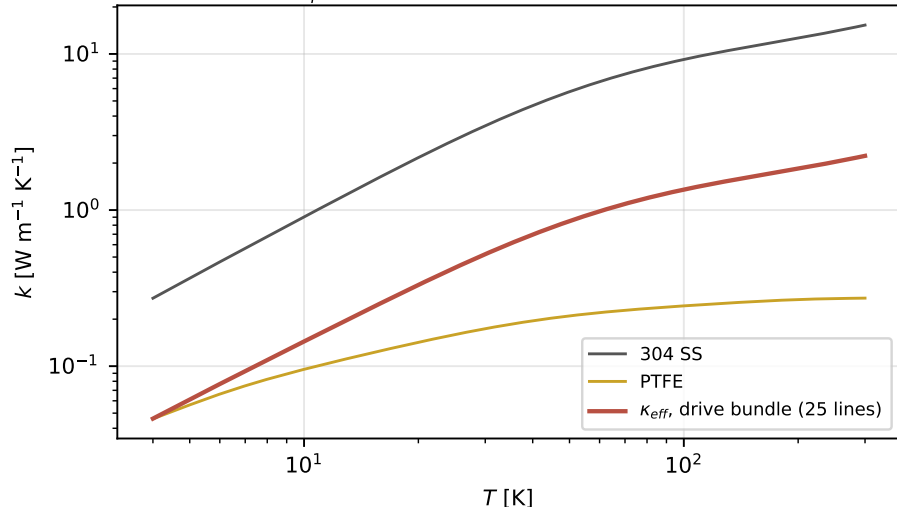


Figure 2: Harness representation: (a) the 65 coaxial lines in four functional families, each lumped into one 20 mm cylinder; (b) the UT-085 cable cross-section with the component areas of Eq. (6); (c) the component conductivities and the resulting effective conductivity of the drive bundle.

Table 1: Enclosure model: geometry, boundary conditions and numerical settings.

Domain	$0.718 \times 0.718 \times 0.335$ m; OVC and shield as snapped stereolithography (STL) surfaces; four coax cylinders $\varnothing 20$ mm at radius 0.15 m
Regions	domain0 (OVC–shield gap), domain1 (warm plate–cold plate/shield gap), coax.L1..L4 (solids)
Gas in gaps	vacuum: $p = 10^{-3}$ Pa, $g = 0$, $\mu = 10^{-11}$ Pa s (no convection, negligible gas conduction; the residual-gas bound is free-molecular helium at 10^{-4} Pa, 24 mW or 2.3 % of the harness conduction, Supplementary S1)
Temperatures	fixed: warm plate and OVC at T_{hot} ; shield and cold plate at T_{cold} ; cases (300, 50), (300, 77), (50, 4), (77, 4) K
Emissivity (baseline)	warm surfaces 0.04 (polished Al); shield and cold plate 0.02 (polished Cu-plated steel); harness 0.02
Emissivity (variants)	± 50 % (0.02/0.01 and 0.06/0.03); multi-layer-insulation (MLI) equivalent 0.003 on all first-stage surfaces
Radiation	viewFactor , grey diffuse, one band, solved every 10 iterations; agglomeration 250 faces per patch at the coarsest level (verification at 400 and 800)
Harness contact	bundles fixed at T_{hot} at the warm cap and T_{cold} at the cold cap; no contact resistance; lateral surfaces exchange by radiation only
Solver	chtMultiRegionSimpleFoam , semi-implicit method for pressure-linked equations (SIMPLE), residuals $< 10^{-7}$, 2900 iterations from a uniform initial field

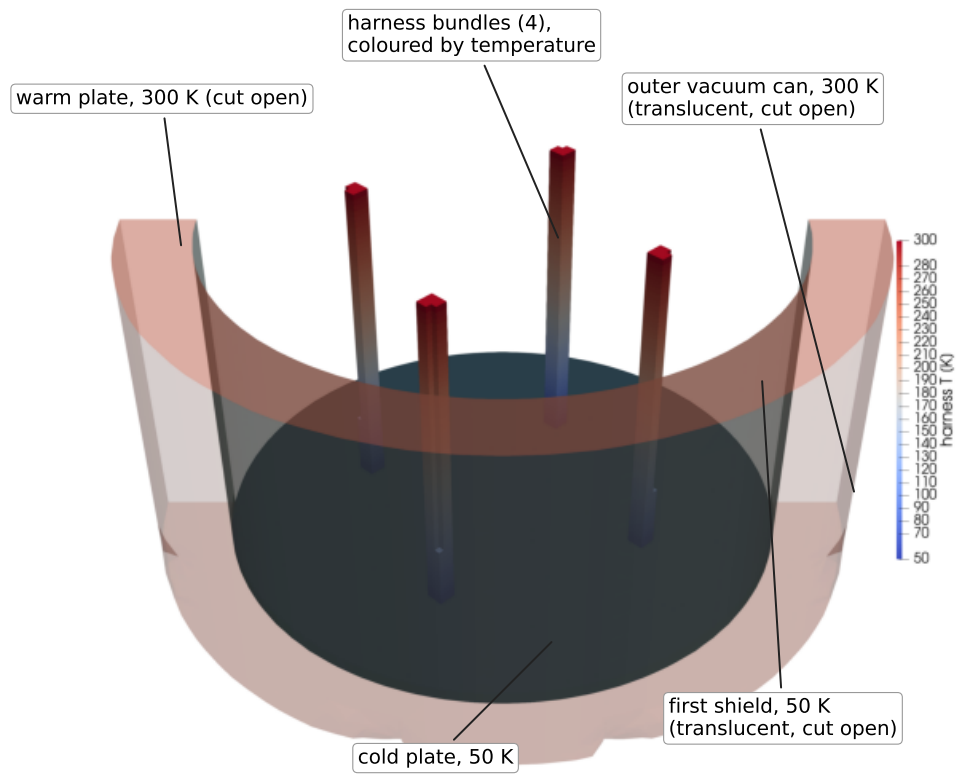


Figure 3: Cut-away of the enclosure CHT domain for the $300 \rightarrow 50$ K case. The warm plate, outer vacuum can and first shield are cut open on the near side; the four harness bundles, coloured by their computed temperature, span the gap from the warm plate to the cold plate. The vacuum gaps between the surfaces carry radiation only.

view-factor matrix is built on agglomerated faces, the radiative load was re-computed at 250, 400 and 800 coarse faces per patch and its closure $\sum_j F_{ij}$ monitored (Supplementary Table S5). The emissivity of polished cryogenic surfaces is uncertain by roughly a factor of 1.5 [37]; the $\pm 50\%$ runs give the uncertainty band carried through the ledger. An MLI-equivalent variant with $\varepsilon = 0.003$ on the first-stage surfaces represents a 10–20 layer blanket [38, 1].

What the three-dimensional model adds.. For a parallel-plate enclosure with fixed wall temperatures, the stage-integrated radiative and conductive loads are analytic, and the CHT model must reproduce them; that agreement is the model’s verification, not its result. The model contributes what the closed forms cannot: the distribution of radiative flux over the shield (which sets local shield temperatures once a real, finitely conductive shield is considered), the coupling between the harness and the surrounding radiation field, the first-stage duty that sets the cold-plate load, and the dependence of the 4 K-stage load on the first-stage temperature that closes the loop between the cold-plate and ledger models (Section 5). The partition of conduction among harness families follows from the cable counts and is not claimed as a result. For the cold plate the closed forms are the fully developed duct limits and the Gnielinski correlation; what the model adds there is the developing-flow Nusselt map across Reynolds number and pressure, the departure of a single-wall-heated duct from the correlations for uniformly heated ducts, and the wall-temperature distributions, including the local maximum that sets the saturation margin, none of which a correlation supplies (Section 4).

2.3. Cold-plate heat-exchanger model (Q2)

Basis of comparison.. The two fluids differ in phase, temperature, density by a factor of 800 and thermal conductivity by a factor of 3.1, so no single common basis is neutral. Three bases are therefore reported: matched Reynolds number (same flow regime), matched mass flow, and matched pumping power. All runs remove the same duty of 9.4 W, the conservative first-stage load of Section 3. Helium enters at 45 K, at 1, 5 or 18 bar (the ITER shield loop runs at 18 bar [4]); nitrogen enters as a liquid at 77 K in a 5 bar loop, for which $T_{\text{sat}} = 94.0$ K (87.9 K at 3 bar). The Boussinesq solution does not depend on the loop pressure, since the 77 K liquid properties change by less than 0.3 % between 3 and 5 bar, so one set of fields serves both pressures; the saturation margin $T_{\text{sat}} - T_{\text{w,max}}$, with $T_{\text{w,max}}$ the hottest face

of the heated-wall patch, is reported for every run to confirm single-phase operation. Table 2 lists the operating quantities and Fig. 4 the geometry.

Property reference state.. The helium transport properties are the NIST values at 50 K and 1 bar, $\mu = 6.36 \mu\text{Pa s}$ and $k = 0.0467 \text{ W m}^{-1} \text{ K}^{-1}$ ($\text{Pr} = 0.709$), held constant in each run with the density from the ideal-gas law; the inlet is 45 K and the wall reaches 106 K at $Re = 500$, over which range μ and k both rise by about 65 %. Two sensitivity runs with cubic $\mu(T)$ and $k(T)$ fitted to the NIST data over 40–100 K (fit error below 0.05 %) are reported in Section 4.1. The reference state matters: gas properties taken 35 K too cold would understate h by a factor of about 2.5.

Metrics.. The heat-transfer coefficient is $h = \dot{Q}/[A_{\text{heated}}(T_w - T_b)]$ with T_w the area-averaged heated-wall temperature and T_b the mean of the inlet and the mass-flux-weighted outlet temperature; $Nu = hD_h/k$; the Darcy friction factor $f = 2\Delta p D_h/(\rho U^2 L)$; pumping power $\dot{W}_p = \dot{m}\Delta p/\rho$; $\text{NTU} = hA/(\dot{m}c_p)$ and $\varepsilon = 1 - e^{-\text{NTU}}$ [39]; the Mouromtseff number $\text{Mo} = k^{0.6}\rho^{0.8}c_p^{0.4}\mu^{-0.4}$ [40]. Laminar results are checked against the fully developed square-duct limits for one heated and three adiabatic walls, $Nu_\infty \approx 2.71$ [41] (a finite-difference solution of the fully developed problem on the present cross-section gives 2.69; the four-wall value is 3.61) and $fRe = 56.9$, and turbulent results are compared with the Gnielinski correlation [42], which is for uniformly heated ducts and serves as a reference rather than a benchmark. Wall temperatures are taken on the heated-wall patch itself: its area mean, its hottest and coldest faces, and the face-value profile along the centre row of the first channel.

Grid convergence.. Three systematically refined meshes (refinement ratio 1.5) were run for both fluids at $Re = 2300$ and the discretisation uncertainty expressed as a GCI [43] (Section 4). For the turbulent runs the first-cell y^+ on the graded mesh averages 2.5–5.6 with a maximum of 9.5, within the range handled by the low-Reynolds wall treatment of the SST model.

2.4. Work ledger ($Q3$)

A load $\dot{Q}_i$ absorbed at T_i and rejected at $T_0 = 300 \text{ K}$ costs

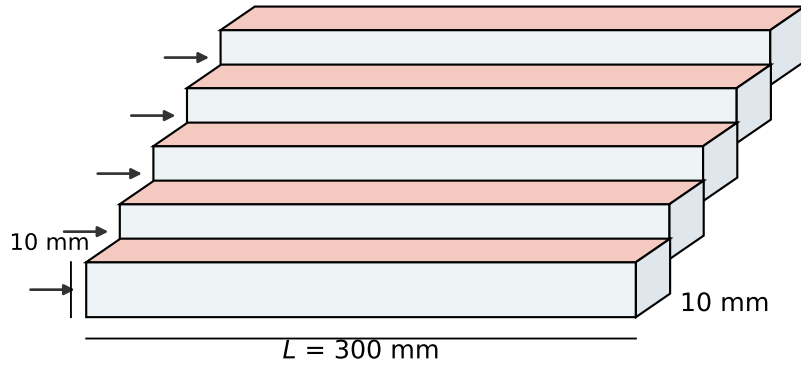
$$\dot{W}_i = \frac{\dot{Q}_i}{\eta_i} \frac{T_0 - T_i}{T_i}, \quad (7)$$

Table 2: Cold-plate model: geometry, coolant states, properties and numerics. Properties from the NIST reference equations of state via CoolProp [32], evaluated at the inlet state and held constant except for the helium density (ideal gas) and the nitrogen buoyancy term.

Shared geometry and duty		
Channels	five parallel square ducts, one wall heated	
Dimensions	$D_h = 10 \text{ mm}$, $L = 0.30 \text{ m}$; $A_{\text{heated}} = 0.015 \text{ m}^2$, $A_{\text{cs}} = 5 \times 10^{-4} \text{ m}^2$	
Duty	9.4 W, uniform flux $q_w = 627 \text{ W m}^{-2}$ on the heated wall	
Mesh	$180 \times 18 \times 18$ cells per duct, wall-graded (ratio 6); 2.92×10^5 cells; GCI on 86k / 292k / 984k	
Gravity	off (forced convection); on for three sensitivity runs	
	Helium gas	Liquid nitrogen
<i>Coolant state</i>		
Inlet temperature	45 K	77 K
Loop pressure	1, 5 or 18 bar	5 bar ($T_{\text{sat}} = 94.0 \text{ K}$; 87.9 K at 3 bar)
Phase	ideal gas, variable density	single-phase liquid (margin reported)
<i>Properties at the inlet state</i>		
Density ρ [kg m^{-3}]	0.961 / 4.80 / 17.3 at 1 / 5 / 18 bar	808
c_p [$\text{J kg}^{-1} \text{K}^{-1}$]	5201	2038
Viscosity μ [$\mu\text{Pa s}$]	6.36	163.5
k [$\text{W m}^{-1} \text{K}^{-1}$]	0.0467	0.1457
Prandtl number Pr	0.71	2.29
β [K^{-1}]	$1/T$ (ideal gas)	5.6×10^{-3}
<i>Flow range</i>		
Reynolds numbers	500, 1000, 1500, 2300 (laminar); 5000, 10 000 (k - ω SST)	
$\dot{m}$ at $Re = 2300$ [kg s^{-1}]	7.31×10^{-4}	1.88×10^{-2}
U at $Re = 2300$ [m s^{-1}]	1.52 (1 bar); 0.084 (18 bar)	0.047
<i>Numerics</i>		
Solver	buoyantSimpleFoam (ideal gas)	buoyantBoussinesqSimpleFoam
Convergence	residuals $< 10^{-6}$	residuals $< 10^{-7}$
Energy closure	per run; $< 0.2 \%$ for every forced-convection case	

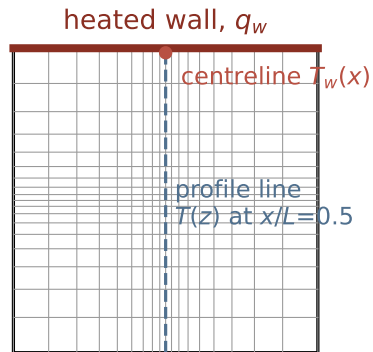
(a) cold-plate heat-exchanger domain

heated wall (top of every duct): uniform $q_w = \dot{Q}/A_{heated} = 627 \text{ W m}^{-2}$



inlet: T_{in} , $\dot{m}$ (He: mass-flow BC; LN₂: velocity BC); outlet: fixed p
 other walls adiabatic and no slip; five square ducts, $D_h = 10 \text{ mm}$, pitch 18 mm
 heated area 0.015 m^2 (top walls), flow area $5 \times 10^{-4} \text{ m}^2$

(b) duct cross-section: mesh and sampling lines



18×18 cells per duct
 two-sided grading, ratio 6
 first cell 0.19 mm
 180 cells along L
 2.9×10^5 cells in total

Figure 4: Cold-plate heat-exchanger model: (a) the five heated square ducts with boundary conditions and dimensions; (b) the graded duct cross-section with the two sampling lines used for the wall-temperature profiles.

198 where η_i is the stage efficiency as a fraction of Carnot. The ideal ledger sets
 199 $\eta_i = 1$: the multiplier is 5.0 at 50 K, 2.9 at 77 K and 74 at 4 K. The real ledger
 200 uses published hardware. A commercial two-stage PTR (Cryomech PT415)
 201 delivers 1.5 W at 4.2 K together with 40 W at 45 K for 10.7 kW of input [33],
 202 so the two stages together return 3.1 % of their combined Carnot work. A
 203 single machine cannot separate the two stage efficiencies, so the ledger takes
 204 them from Radebaugh’s survey [30]: $\eta_2 = 0.008$ – 0.015 at 4 K and $\eta_1 = 0.10$ –
 205 0.15 at 45–80 K (Supplementary S3). These ranges are consistent with the
 206 PT415: at their midpoints the two stages would need 11.4 kW, within 7 % of
 207 the rated input. Liquid nitrogen carries 199 kJ kg^{-1} of latent heat at 1 bar;
 208 its ideal production work is $0.213 \text{ kWh kg}^{-1}$ [2] and real air-separation plants
 209 need 0.5 – 0.7 kWh kg^{-1} [31], an exergy efficiency $\eta_{\text{N}_2} \approx 0.3$ – 0.4 . The 4 K stage
 210 of the nitrogen route is still served by a cryocooler, so its η_2 is the same in
 211 both routes. The ranges above are propagated as bands.

212 **3. Results Q1: the passive penalty of a 77 K shield**

213 *3.1. Radiation verification and the view-factor closure*

214 The first-stage radiative load computed with the view-factor matrix as
 215 generated does not converge under agglomeration refinement: 9.46, 8.41, 6.49
 216 and 3.26 W at 60, 250, 400 and 800 coarse faces per patch, against 7.07 W
 217 from the parallel-plate closed form (Fig. 5). The cause is closure. In a closed
 218 enclosure every row of the view-factor matrix must sum to one; the generated
 219 rows sum to between 0.50 and 1.13 (mean 0.97–1.01), and finer agglomeration
 220 worsens the scatter. With surface emissivities of 0.02–0.04 the radiosity is
 221 reflected of order $1/\varepsilon$ times before it is absorbed, so a per-reflection leak of a
 222 few per cent compounds into a load error of tens of per cent. Normalising each
 223 row of the generated matrix to unity removes the artefact: at the production
 224 level of 250 faces the $300 \rightarrow 50 \text{ K}$ load becomes 7.36 W (4.1 % above the closed
 225 form), the $300 \rightarrow 77 \text{ K}$ load 7.34 W (4.2 %), and the two 4 K-stage loads 3.16
 226 and 17.8 mW (0.5 % above 3.15 and 17.7 mW). The remaining 3–4 % at the
 227 first stage, which shrinks with refinement, combines the geometric departure
 228 of the real enclosure, with its harness penetrations, from the parallel-plate
 229 idealisation with the residual agglomeration error. All loads below are the
 230 row-normalised values; the unnormalised production values are retained as an
 231 upper band. Row normalisation enforces closure but not reciprocity, $A_i F_{ij} =$
 232 $A_j F_{ji}$; the least-squares rectification procedures compared by Taylor and
 233 Luck [44] enforce both and would be the next refinement. The corrected

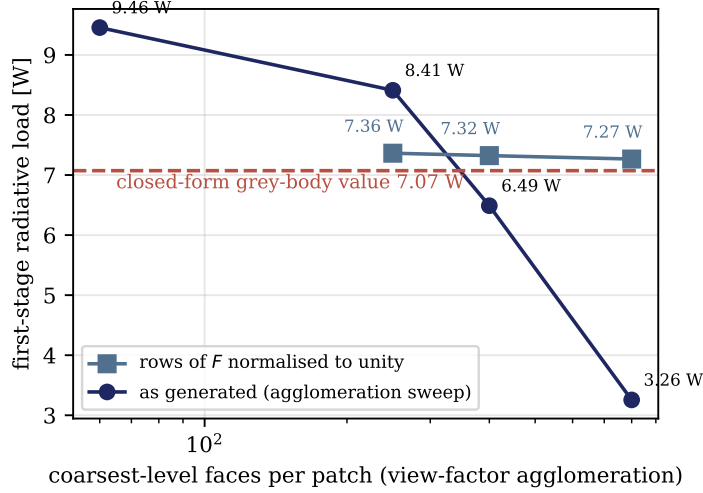


Figure 5: First-stage radiative load (300 $\rightarrow$ 50 K) against the view-factor agglomeration level with the matrix as generated (circles) and with its rows normalised to unity (squares), compared with the closed-form grey-body value. The monotonic collapse of the generated matrix under refinement is a closure artefact amplified by the low emissivities; the normalised load converges toward the closed form.

method converges under agglomeration refinement: the normalised load is 7.36 W at 250 faces, 7.32 W at 400 and 7.27 W at 800, 4.1, 3.6 and 2.8 % above the closed form, where the uncorrected matrix gave 8.41, 6.49 and 3.26 W. Two further checks quantify what normalisation does and does not repair. Global conservation, the net radiative power summed over every surface of a closed region, must vanish: the generated matrices leave 1.5 W unbalanced at 400 faces and 14 W at 800, the normalised ones 0.10, 0.13 and 0.36 W at 250, 400 and 800 faces (1.3, 1.8 and 4.9 % of the load). Reciprocity, which normalisation does not enforce, was audited with the coarse-face areas rebuilt from the mesh (script `reciprocity.py`): the area-weighted error $\sum |A_i F_{ij} - A_j F_{ji}| / \sum A_i F_{ij}$ of the normalised matrices is 3.9, 5.5 and 12.6 % at 250, 400 and 800 faces in the outer gap region. Both residuals grow with refinement, which is why the production level is 250 faces and why the 3–4 % gap to the closed form is reported as the accuracy of the method rather than refined away.

Table 3: Stage loads by mechanism (row-normalised view factors; closed-form value in parentheses), with the unnormalised production value and the $\pm 50\%$ emissivity band as upper and lower bounds.

Case	Radiation	Conduction	Total	Prod.	$\varepsilon \pm 50\%$ [W]
300 $\rightarrow$ 50 K	7.36 W (7.07)	1.029 W (1.029)	8.39 W	9.44 W	6.5–12.5
300 $\rightarrow$ 77 K	7.34 W (7.04)	0.962 W (0.962)	8.30 W	9.35 W	6.4–12.4
50 $\rightarrow$ 4 K	3.16 mW (3.15)	70.3 mW (70.3)	73.5 mW	75.6 mW	–
77 $\rightarrow$ 4 K	17.8 mW (17.7)	162.1 mW (162.1)	179.9 mW	191.8 mW	–
300 $\rightarrow$ 50 K MLI	1.69 W (1.47)	1.029 W	2.72 W	2.84 W	–
300 $\rightarrow$ 77 K MLI	1.68 W (1.46)	0.962 W	2.65 W	2.77 W	–

3.2. Stage loads and mechanism split

At the first stage radiation dominates and the shield temperature barely matters (Table 3, Fig. 6). The radiative flux scales with $T_{\text{hot}}^4 - T_{\text{cold}}^4$ and 300⁴ dwarfs both 50⁴ and 77⁴, so the 77 K shield intercepts 99.6% of what the 50 K shield intercepts. Harness conduction falls with the smaller span, 962 against 1029 mW. The first-stage loads are 8.39 W (50 K) and 8.30 W (77 K); the emissivity band of $\pm 50\%$ moves them between 6.5 and 12.5 W but leaves the 1% difference between routes unchanged. The conservative production value of 9.4 W is used as the cold-plate duty in Section 4.

At the 4 K stage the picture inverts. Residual radiation rises by the factor $(77^4 - 4^4)/(50^4 - 4^4) = 5.6$, but from a base of 3 mW; harness conduction rises by the factor 2.31 from a base of 70 mW. The 77 K route therefore delivers 106 mW more to the 4 K plate: 92 mW of conduction and 15 mW of radiation, 86% conductive. Through the ideal 4 K multiplier of 74 this is 7.9 W of room-temperature power, 6.8 W of it from the harness. Every stage-integrated ratio agrees with its closed form to better than 0.5%.

The MLI-equivalent shield ($\varepsilon = 0.003$ on the first-stage surfaces) cuts the first-stage radiation to 1.69 W (50 K) and 1.68 W (77 K) with the row-normalised matrix (closed form 1.47 and 1.46 W; the unnormalised production matrix gives 1.81 W on either route, the same +23% closure bias as at the bare-shield level), so the first-stage loads fall to 2.72 and 2.65 W, with conduction's share rising to 36–38%. The 4 K-stage loads are unaffected. This variant is the one that changes the ledger verdict in Section 5.

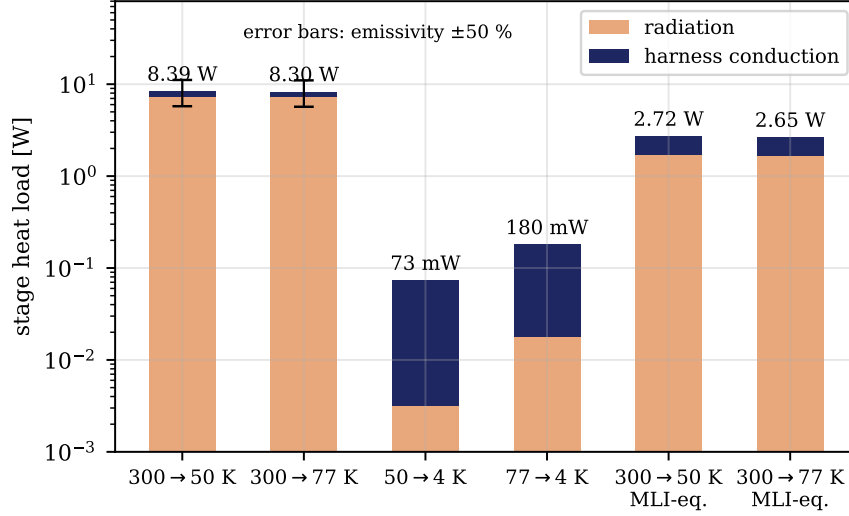


Figure 6: Stage loads for the four enclosure cases and the two MLI-equivalent variants (row-normalised view factors, the values used in the text), decomposed by mechanism, with the $\pm 50\%$ emissivity band. The first stage is radiation-dominated and nearly identical between routes; the 4 K stage is conduction-dominated and is where the routes diverge.

272 3.3. Spatial distribution

273 The view-factor solution concentrates the radiative flux on the shield
 274 regions with a direct line of sight to the OVC ($8.2\text{--}10.8\text{ W m}^{-2}$ across the
 275 shield) and leaves the harness penetrations cooler (Fig. 7). Conduction is
 276 shared among the bundles in the ratio of their line counts: on the $77 \rightarrow 4\text{ K}$
 277 span the drive and flux families deliver 125 of the 162 mW. These two bundles
 278 are the targets for any intermediate heat sink, and the harness conduction
 279 integral, not the cryogen, is the lever on the 4 K-stage penalty.

280 4. Results Q2: helium gas versus liquid nitrogen in the cold plate

281 4.1. Verification

282 Every forced-convection run closes its energy balance, wall heat input
 283 against mass-flux-weighted enthalpy rise, to within 0.2% (Supplementary
 284 Table S7). At $Re = 500$ the helium flow is thermally developing over the
 285 first 59% of the channel ($L_t/L = 0.59$) and returns $Nu = 3.66$, 35% above
 286 the fully developed limit of 2.71 for a square duct heated on one wall [41],
 287 the excess being the developing-length contribution; every other laminar run

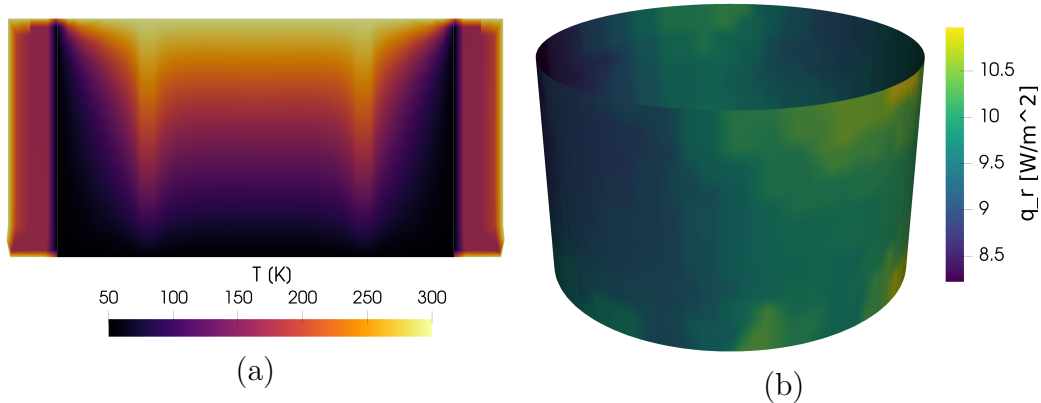


Figure 7: (a) Mid-plane temperature and (b) shield radiative-flux map of the 300 → 50 K case.

288 has a longer development length (L_t/L up to 8.8 for nitrogen) and lies fur-
 289 ther above the limit, as it must. The three-grid studies (Table 4) converge
 290 monotonically for both fluids: the production-mesh grid convergence index
 291 is 0.7 % (Nu) and 2.5 % (f) for helium and 2.2 % and 1.6 % for nitrogen. In
 292 the turbulent runs the first-cell y^+ on the heated wall averages 2.5–5.6 with
 293 a maximum of 9.5, within the low-Reynolds treatment of the SST model.
 294 Nitrogen’s turbulent Nusselt numbers lie 8–9 % below the Gnielinski correla-
 295 tion [42] and helium’s 28–32 % below it. The correlation is for a duct heated
 296 on all walls; in the laminar limit one heated wall lowers Nu by 25 % (2.71
 297 against 3.61), and the larger deficit of the low-Prandtl gas, whose thermal
 298 layer fills the duct, is consistent with that boundary condition but is not
 299 separately verified here, so the correlation is a reference, not a benchmark,
 300 for these runs.

301 *Property sensitivity.* Two helium runs at 1 bar, $Re = 500$ and 2300, were
 302 repeated with cubic $\mu(T)$ and $k(T)$ fitted to the NIST data over 40–100 K
 303 (Section 2.3) and a heat-flux wall condition that uses the local conductivity
 304 (delivered 9.399 W, closure 0.01 %). Because k rises with temperature, the
 305 hot near-wall gas conducts better than the 50 K reference value: at $Re = 500$
 306 the Nusselt number rises from 3.66 to 4.16 (+14 %), the plate mean falls from
 307 87.4 to 83.0 K and the hottest face from 106.1 to 97.3 K, while the friction
 308 factor and pumping power rise by 5 %; at $Re = 2300$ the changes are +5 % in
 309 Nu , -1.0 K in the mean and -2.9 K in the hottest face (79.7 to 76.9 K), with
 310 the pumping power unchanged. The 0.5 K by which that hottest face clears

Table 4: Grid convergence at $Re = 2300$ (Celik et al. [43]; meshes of 86k, 292k and 984k cells, refinement ratio 1.5). The production mesh is the medium one.

Fluid	Quantity	Coarse	Medium	Fine	p	Extrapolated	GCI_{medium}
He (1 bar)	Nu	6.439	6.449	6.457	0.64	6.49	0.7 %
	f	0.0632	0.0642	0.0647	1.40	0.0655	2.5 %
LN ₂ (3 bar)	Nu	10.51	10.44	10.38	0.87	10.26	2.2 %
	f	0.0592	0.0599	0.0603	1.71	0.0607	1.6 %

the 77.4 K limit is within the uncertainty of the quantity: the three-grid study moves it by 0.17 K (79.85, 79.74 and 79.68 K), and the property fits and the constant c_p add a few tenths, so the hottest-face optimum of Section 4.2 is stated as a range, $Re \approx 2300$ to 5000 with a pumping-power ratio of 4 to 30, rather than as a point. The constant-property helium results are therefore conservative toward helium by up to 14 % in h at the lowest Reynolds number and by 5 % at the design point; the rankings of Sections 4.2–4.4 are not affected, and the one statement they touch, the hottest-face optimum, is qualified where it is made. Both runs are listed in Supplementary Table S7.

4.2. Operating maps

Figure 8 maps the two coolants over $Re = 500$ – 10^4 at the fixed 9.4 W duty. Nitrogen’s heat-transfer coefficient exceeds helium’s by a factor of 4.9–5.1 throughout the laminar range and 6.2–6.3 in the turbulent range; the laminar ratio is set by the conductivity ratio of 3.1 times the ratio of developing-flow Nusselt numbers, which favours the high-Prandtl liquid. The consequence for the plate is stark. At atmospheric pressure the helium-cooled wall averages 87 K at $Re = 500$ and 67 K at $Re = 2300$, with hottest faces at the downstream end of the channel of 106 and 80 K (Fig. 8d). The plate mean falls below the 77 K nitrogen temperature above $Re \approx 970$, but the hottest face does so only in turbulent flow, at $Re = 5000$ (61 K maximum, 55 K mean); the nominal 50 K of the helium route is approached only at $Re = 10^4$ and 65 mW of pumping power (50.3 K mean, 54.3 K maximum). The nitrogen-cooled wall stays between 78 and 85 K in the mean and below 89.2 K at its hottest face over the whole map. Its saturation margin at 5 bar is 4.8 K at $Re = 500$, 9.9 K at 2300 and 15.4 K at 10^4 . At 3 bar the $Re = 500$ run would exceed saturation locally (89.2 against 87.9 K) and is two-phase at the downstream end, and at 1 bar the wall would exceed the 77.4 K saturation temperature in every run, so a bath-fed loop at atmospheric

339 pressure would boil on the plate at this duty and a pressurised or subcooled
 340 loop is a requirement, not a refinement.

341 Pumping power separates the fluids in the opposite sense once pressure
 342 enters. At 1 bar helium needs 38 times nitrogen’s pumping power at matched
 343 Reynolds number (1.39 against 0.037 mW at $Re = 2300$), because its mass
 344 flow is 26 times smaller but its velocity 32 times larger. Raising the helium
 345 pressure at fixed Reynolds number leaves the mass flow, the Nusselt number
 346 and the wall temperature unchanged to four significant figures and reduces
 347 the pumping power as the inverse square of the density: 25.0 times at 5 bar
 348 and 324 times at 18 bar (Fig. 8c). At the 18 bar of the ITER and DTT shield
 349 loops [4, 5] helium’s pumping power is $4.3 \mu\text{W}$, 8.5 times below nitrogen’s; the
 350 parallel-path gaseous-helium shields of LHD operate in the same regime [6].

351 *Optimum operating points..* The maps answer the second half of Q2 once
 352 an optimum is defined: the minimum pumping power at which the plate
 353 meets a wall-temperature limit. For helium the natural limit is 77.4 K, the
 354 temperature a nitrogen-bath plate would hold, since a warmer helium plate
 355 forfeits the 50/77 K distinction of the passive comparison; for nitrogen it is
 356 the 5 bar saturation temperature less a 2 K single-phase margin, 92.0 K. The
 357 answer depends on which wall temperature is held to the limit. If the hottest
 358 face must comply, no laminar helium run of the constant-property sweep
 359 does: the 1 bar laminar curve of Fig. 8d stays above 79.7 K up to $Re = 2300$,
 360 and the first compliant run is the turbulent $Re = 5000$ case, at 9.95 mW
 361 of pumping power at 1 bar, 0.40 mW at 5 bar and $31 \mu\text{W}$ at 18 bar (the gap
 362 between the laminar and turbulent runs is not interpolated). The margin
 363 at $Re = 2300$ is small, 2.4 K, and the variable-property run of Section 4.1
 364 closes it (76.9 K), so with the property variation resolved the hottest-face
 365 optimum moves to $Re \approx 2300$ at 1.39 mW (1 bar) or $4.3 \mu\text{W}$ (18 bar), four
 366 times nitrogen’s rather than thirty. Nitrogen complies at $Re = 500$, the
 367 lowest run in the sweep, with $1.0 \mu\text{W}$. If the plate mean is the criterion,
 368 helium crosses 77.4 K at $Re \approx 970$, where the pumping power is 0.20 mW
 369 at 1 bar and $0.62 \mu\text{W}$ at 18 bar, and nitrogen again at $Re = 500$ (stars in
 370 Fig. 8c,d). Compared at their optima, pressurised helium therefore needs 4
 371 to 30 times nitrogen’s pumping power on the hottest-face criterion and 40 %
 372 less on the plate-mean criterion. Both are microwatts and neither enters
 373 the ledger of Section 5; what the comparison establishes is that the 38-fold
 374 penalty of atmospheric helium at matched Re is the cost of running the gas
 375 loop at the wrong pressure rather than a property of the gas, and that a

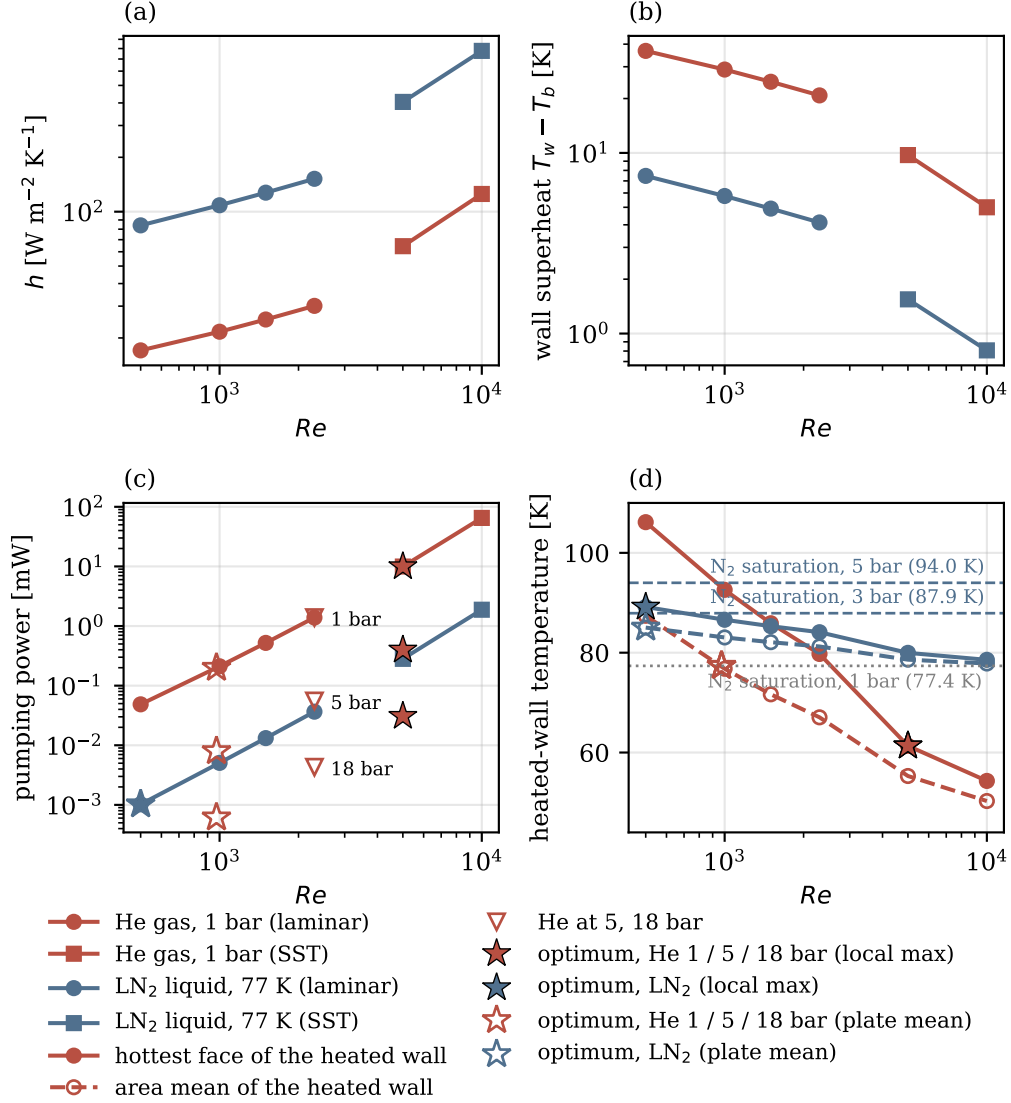


Figure 8: Operating maps at 9.4 W duty: (a) heat-transfer coefficient, (b) wall superheat, (c) pumping power with the helium pressure series at $Re = 2300$, and (d) heated-wall temperature, hottest face (filled) and area mean (open), with the nitrogen saturation temperatures at 1, 3 and 5 bar. Circles: laminar; squares: k - ω SST; stars: each fluid's optimum, the minimum pumping power at which the wall temperature meets its limit (77.4 K for helium, 92.0 K for nitrogen), filled for the hottest-face criterion and open for the plate-mean criterion.

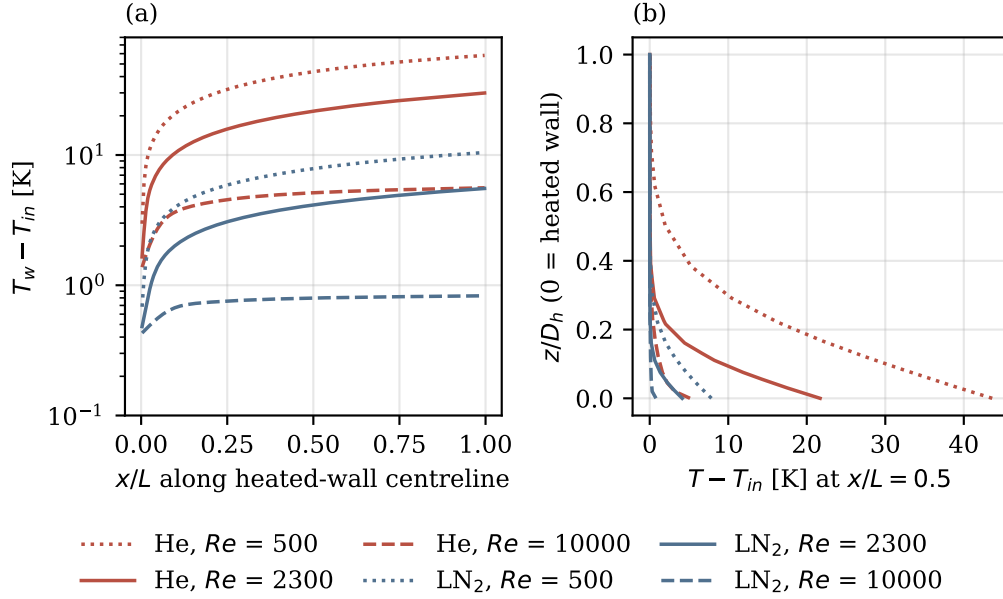


Figure 9: (a) Heated-wall centreline temperature rise above the inlet and (b) cross-channel temperature profile at $x/L = 0.5$ for both fluids at $Re = 500$, 2300 and 10^4 .

helium plate must be checked against its hottest face, 13 K above the mean at $Re = 2300$, not against the mean that sets h .

4.3. Wall-temperature distributions

Figure 9 gives the heated-wall temperature along the centre row of the first channel, read from the wall faces themselves, and the cross-channel profile at mid-length. Along the wall the temperature rise grows as $x^{1/3}$ in the developing region and then linearly; at the outlet the helium wall sits 30 K above its inlet at $Re = 2300$ and 58 K at $Re = 500$, nitrogen 5.5 and 10.5 K, and the hottest faces of the patch, near the duct side walls at the outlet where the velocity is lowest, are a further 4.8 K (helium) and 1.6 K (nitrogen) warmer at $Re = 2300$. Across the channel the helium thermal layer at $Re = 500$ fills the duct, whereas nitrogen's occupies the near-wall 30 % even at $Re = 500$. This is the geometric origin of the heat-transfer ratio and of the plate temperatures in Fig. 8d.

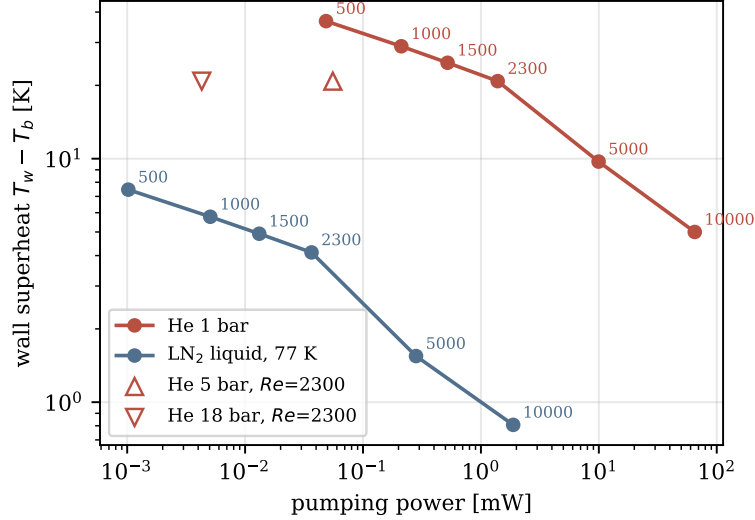


Figure 10: Wall superheat against pumping power for both fluids over the Reynolds sweep (labels: Re), with the helium pressure series at $Re = 2300$.

4.4. Ranking on three bases

Table 5 collects the design point and Fig. 10 the trade-off between wall superheat and pumping power. The ranking depends on the basis. At matched Reynolds number nitrogen wins heat transfer by a factor of 5 and pumping power by 38; the effectiveness-NTU pair favours helium (0.11 against 0.06), a capacity-rate artefact of the gas's small $\dot{m}c_p$ that says nothing about how cold the plate is. At matched mass flow nitrogen's Reynolds number would fall to about 90, below the sweep; the fully developed one-wall limit $Nu_\infty \approx 2.71$ bounds its heat-transfer coefficient from below at about $40 \text{ W m}^{-2}\text{K}^{-1}$, 1.3 times helium's at $Re = 2300$, but this basis is not computed and is quoted only as a bound. At matched pumping power, the basis a plant designer pays for, nitrogen at the pumping power of 1 bar helium ($Re = 2300$) holds a superheat of 0.9 K against helium's 20.8 K, a factor of 23; at the pumping power of 18 bar helium the factor is 3.5 (5.9 against 20.8 K). In the turbulent range the curves in Fig. 10 are parallel: a given superheat costs helium at 1 bar roughly 30 times the pumping power it costs nitrogen, and the 18 bar helium point sits inside the nitrogen envelope on pumping power alone but outside it on superheat. No single number ranks the coolants; the operating point and the pressure do.

Table 5: Design point, $Re = 2300$, 9.4 W, forced convection. He at 1 bar (18 bar in parentheses where pressure matters); LN₂ at 5 bar (3 bar in parentheses for the saturation margin). GCI on Nu and h : 0.7 % (He), 2.2 % (LN₂).

Quantity	He, 50 K	LN ₂ , 77 K	LN ₂ /He
Mass flow $\dot{m}$ [g s ⁻¹]	0.731	18.8	25.7
Volumetric flow [cm ³ s ⁻¹]	702 (39)	23.3	0.033
Mean velocity U [m s ⁻¹]	1.52 (0.084)	0.047	0.031
Inlet / outlet temperature [K]	45.0 / 47.5	77.0 / 77.2	–
Pressure drop Δp [Pa]	1.98 (0.11)	1.57	0.80
Pumping power $\dot{W}_p$ [μ W]	1388 (4.3)	36.6	0.026 (8.5)
Wall temperature, mean / hottest face ^a [K]	67.1 / 79.7	81.2 / 84.1	–
Wall superheat $T_w - T_b$ [K]	20.8	4.12	0.20
Heat-transfer coefficient h [W m ⁻² K ⁻¹]	30.1	152	5.05
Nusselt number Nu	6.45	10.4	1.62
Darcy friction factor f	0.064	0.060	0.93
NTU / effectiveness ε	0.119 / 0.112	0.060 / 0.058	0.50 / 0.52
Mouromtseff number (SI units)	567 (5720)	4.6×10^4	81 (8)
Buoyancy parameter Gr/Re^2	0.02	1.05	–
Saturation margin $T_{sat} - T_{w,max}$ [K]	–	9.9 (3.8)	–
Energy-balance closure	0.06 %	0.001 %	–

^apatch area mean / hottest patch face (surfaceFieldValue), at the outlet end.

4.5. Buoyancy sensitivity

The buoyancy parameter Gr/Re^2 of the forced-convection runs is at most 0.63 for helium at 1 bar ($Re = 500$) and above unity for nitrogen in the laminar range (40 at $Re = 500$, 1.0 at 2300), so a real plate in which gravity acts along the heated wall would carry a mixed-convection secondary flow in nitrogen; the one helium case in which orientation would also matter is the dense 18 bar loop, where Gr/Re^2 reaches 7.3 at $Re = 2300$. The four runs with gravity on confirm this and bound its effect. Helium converged and gained 27 % in Nu at $Re = 2300$ and 118 % at $Re = 500$. Nitrogen did not reach a steady state at either Reynolds number: after 10 000 iterations the temperature residual oscillated at 10^{-3} and the energy balance remained open by 11 % and 72 %, the signature of an unsteady buoyant flow that a steady solver cannot represent. The forced-convection nitrogen results therefore describe a plate in which gravity is normal to the heated wall or absent; for a plate with gravity along the wall the steady model gives no bound in either direction, and the ranking of Section 4.4 is stated for forced convection only.

5. Results Q3: the work ledger

Table 7 and Fig. 11 price the two routes with Eq. (7). With ideal multipliers and the bare shield the nitrogen route is cheaper: 37.3 against 47.4 W, a margin of 21 %, because intercepting 8.3 W at a multiplier of 2.9 instead of 5.0 saves 18 W while the 4 K-stage penalty costs 7.9 W. The emissivity band (31.8–49.2 W against 37.8–68.0 W) does not close the gap. With the MLI-equivalent shield the first-stage load shrinks to 2.6–2.7 W, the nitrogen saving to 5.9 W, and the verdict flips: helium 19.0 W against nitrogen 21.0 W. The ideal ledger therefore favours nitrogen where the first-stage load is large and helium where insulation has removed it, which is the regime of the MLI-wrapped shields surveyed by Shu et al. [3] and of the single-layer intercepts of Singh et al. [45].

With hardware efficiencies, and the shields at their nominal temperatures, the verdict is not close in either case. The 4 K stage at 0.8–1.5 % of Carnot turns the 106 mW penalty into 520–980 W, more than the whole first-stage bill of either route, and the helium route costs 840 W (640–1100) against 1300 W (960–1770) for nitrogen with the bare shield and 610 against 1240 W with MLI. On the nitrogen route the first-stage term is not a cryocooler but the liquefaction energy of the liquid evaporated, 0.5–0.7 kWh kg⁻¹ against a latent heat of 199 kJ kg⁻¹, or $M_{N_2} = 9.1$ –12.7 W of plant input per watt

intercepted; this is the basis of Table 7 (a fraction-of-Carnot basis for an 80 K cooler, 7.2–9.7, would lower the nitrogen figures by 1–2 %). The 4 K-stage term is the same cryocooler stage on both routes. Written as a break-even, the nitrogen route is cheaper only if

$$\frac{\eta_2}{\eta_1} \gtrsim \frac{74 \Delta \dot{Q}_2}{5.0 \dot{Q}_{1,\text{He}} - M_{\text{N}_2} \dot{Q}_{1,\text{N}_2} \eta_1}, \quad (8)$$

which with the present loads and the range midpoints $\eta_1 = 0.12$ and $M_{\text{N}_2} = 10.7$ requires a 4 K-stage efficiency of 3.0 % of Carnot for the bare shield and 9 % for the MLI shield, two to ten times what commercial coolers achieve [30]. The lever is $\Delta \dot{Q}_2$: because 86 % of it is harness conduction, a harness that cuts the $77 \rightarrow 4$ K conduction penalty threefold, by heat-sinking the drive and flux bundles or by lower-conductivity outer conductors, brings the bare-shield break-even to 1.3 % of Carnot, within the range of real hardware.

Coupled operating points.. The ledger above holds each shield at its nominal temperature, 50 or 77 K, but Section 4.2 shows that neither cold plate sits there: the helium plate averages 67 K at $Re = 2300$ and reaches 50 K only at $Re = 10^4$, and the nitrogen plate averages 81.2 K at $Re = 2300$ and 77.8 K at 10^4 , 4.2 and 0.8 K above the bath, because the duty must cross the wall superheat. The harness is heat-sunk at the plate, so the 4 K-stage load follows the plate temperature, not the coolant temperature. Table 6 re-evaluates the ledger with the first-stage temperature set to the area-mean plate temperature of the cold-plate model at four operating points, using the harness integral behind Eq. (6) and the T_1^4 scaling of the residual radiation, both of which the enclosure model reproduces to 0.5 %; the helium pumping power is charged at a 50 % circulator efficiency. At matched laminar flow the helium plate’s 67 K costs it 62 mW at the 4 K stage and the nitrogen plate’s 81 K costs 21 mW, and the hardware advantage of helium shrinks from 36 to 13 % (bare shield) and from 51 to 26 % (MLI). When each route runs at the flow its own design implies, helium turbulent at $Re = 10^4$ to hold 50 K and nitrogen laminar at $Re = 2300$, the advantage is 41 and 56 %; with both turbulent it is 36 and 52 %. The ideal ledger moves the other way: at matched laminar flow nitrogen is cheaper by 25 % for the bare shield and by 4 % even with MLI. These are equivalent room-temperature powers under the stated efficiencies and supply energies, not measured plant consumption. What they establish is the sign and the order of the difference, and that

Table 6: Work ledger at coupled operating points: first-stage temperature T_1 equal to the area-mean plate temperature of the cold-plate model (He at 1 bar; 18 bar changes only the negligible pumping term); $\dot{Q}_2$ from the harness integral and the scaled residual radiation; equivalent room-temperature power with ideal multipliers and with mid-range hardware efficiencies (ranges in `data/ledger_coupled.csv`).

Operating point (He / LN ₂)	T_1 [K]	$\dot{Q}_2$ [mW]	$\dot{W}_{\text{ideal}}$ [W]	$\dot{W}_{\text{real}}$ [W]	He advantage,
<i>bare shield</i>					
nominal, decoupled	50 / 77	73 / 180	47.4 / 37.3	840 / 1300	3
matched $Re = 2300$, both laminar	67.1 / 81.2	135 / 201	51.7 / 38.8	1250 / 1450	1
own design points: He $Re = 10^4$, LN ₂ $Re = 2300$	50.3 / 81.2	74 / 201	47.5 / 38.8	850 / 1450	4
both turbulent, $Re = 10^4$	50.3 / 77.8	74 / 184	47.5 / 37.6	850 / 1330	3
<i>MLI-equivalent shield</i>					
nominal, decoupled	50 / 77	73 / 180	19.0 / 21.0	610 / 1240	5
matched $Re = 2300$, both laminar	67.1 / 81.2	135 / 201	23.4 / 22.5	1020 / 1380	2
own design points	50.3 / 81.2	74 / 201	19.1 / 22.5	610 / 1380	5
both turbulent, $Re = 10^4$	50.3 / 77.8	74 / 184	19.1 / 21.3	610 / 1270	5

Table 7: Work ledger at the nominal shield temperatures: stage loads (row-normalised view factors) and equivalent room-temperature power, ideal and hardware-based. Ranges: emissivity $\pm 50\%$ (ideal) and the stage-efficiency and nitrogen supply-energy ranges of Supplementary Table S9 (real).

Shield	Route	$\dot{Q}_1$ [W]	$\dot{Q}_2$ [mW]	$\dot{W}_{\text{ideal}}$ [W]	$\dot{W}_{\text{real}}$ [W]
bare	He, 50 K	8.39	73.5	47.4 (37.8–68.0)	840 (640–1100)
bare	LN ₂ , 77 K	8.30	179.9	37.3 (31.8–49.2)	1300 (960–1770)
MLI-eq.	He, 50 K	2.72	73.5	19.0	610 (450–820)
MLI-eq.	LN ₂ , 77 K	2.65	179.9	21.0	1240 (910–1700)

478 the helium route’s advantage exists only where its plate is actually held near
479 50 K, which the flow rate, not the coolant, decides.

480 6. Discussion

481 *The penalty is a harness property..* The passive cost of the warmer shield is
482 located, sized and attributed: 106 mW at the 4 K stage, 86 % of it conduction
483 along a harness whose drive and flux families carry three quarters. It is
484 a property of the wiring, not of the cryogen, and Eq. (8) shows that, at
485 the efficiencies of present hardware, the wiring is the variable that moves
486 the break-even most. Lumped budgets give the same total, and its split

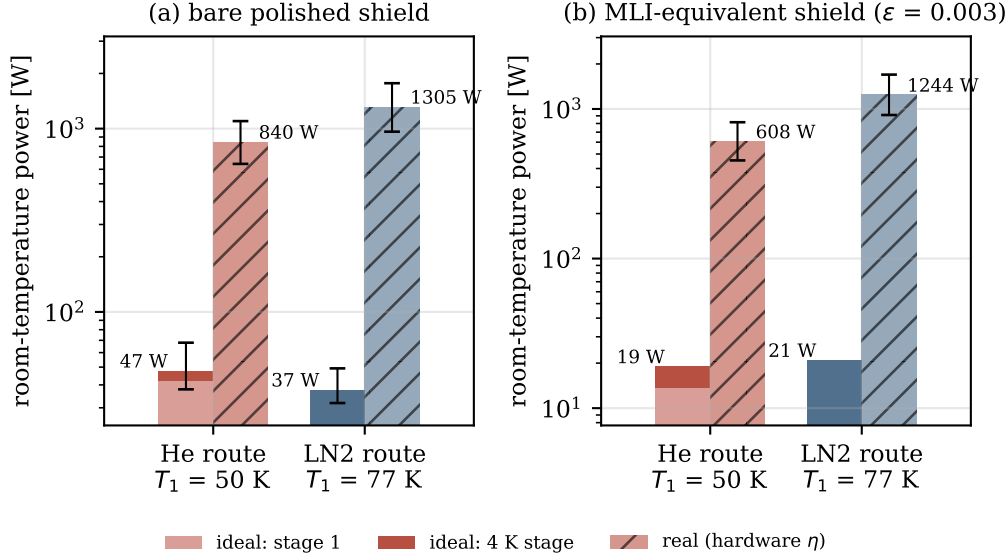


Figure 11: Room-temperature power of the two routes for (a) the bare polished shield and (b) the MLI-equivalent shield: ideal Carnot multipliers split by stage (solid) and hardware efficiencies (hatched), with emissivity and efficiency ranges as error bars.

among bundles follows from the cable counts; what the resolved model adds is the coupling of the harness to the radiation field, the flux distribution over the shield, the closure behaviour of the view-factor matrix, the confirmation that the residual radiation is a minor term at any shield temperature, and the dependence of the 4K load on the plate temperature that the coupled ledger of Table 6 uses. This is the design steer that Li et al.'s optimisation of intercepts on supports and in insulation [11] cannot give for a wired cryostat.

The helium plate floats.. A cold plate cooled by circulated helium gas at atmospheric pressure does not sit at the gas temperature. At the 9.4 W duty it runs 21 K above the bulk at $Re = 2300$, with a hottest face 13 K above that mean; the mean reaches nitrogen temperature at $Re \approx 970$ and the hottest face only at $Re \gtrsim 2300$, and the nominal 50 K is approached only at $Re = 10^4$ and tens of milliwatts of pumping power. Pressurising the loop leaves the wall temperature untouched but makes the pumping power negligible. This is a statement about the circulated-gas architecture and the channel sizing, not about the conductively coupled PTR first stage of Fig. 1a, whose plate temperature is set by the cooler's capacity curve. It does mean that the

504 50/77 K distinction of the passive comparison is only realised on the helium
505 route with an adequate flow or pressure, and Section 4.2 gives the conditions.

506 *The three questions couple through the plate temperature..* The ledger credits
507 the helium route with a 50 K shield, but the cold plate reaches 50 K only at
508 $Re \approx 10^4$; at the laminar design point of $Re = 2300$ and 1 bar its mean is
509 67 K. Evaluating the harness integral of Eq. (6) with the shield at 67 K gives
510 a 4 K-stage conduction of 125 mW, against 70 mW at 50 K and 162 mW at
511 77 K: a helium shield run at that design point forfeits 54 mW, 59 % of the
512 92 mW conduction penalty that separates the routes, and about 4 W of the
513 7.9 W ideal advantage. The 65 mW of pumping power that buys the 50 K
514 plate at $Re = 10^4$ is negligible against that, so on the helium route the flow
515 rate is set by the 4 K ledger and not by the pumping power. The nitrogen
516 plate is not pinned by the bath either: it sits 4.2 K above it at $Re = 2300$
517 and 0.8 K at 10^4 , worth 21 and 4 mW at the 4 K stage, a smaller and flow-
518 insensitive coupling because the liquid’s superheat is small. Table 6 carries
519 both couplings into the ledger, and the operational argument for nitrogen,
520 a plate temperature that the flow rate barely moves, is what remains of the
521 bath-pinning idea.

522 *The basis of comparison decides the ranking..* Matched Reynolds number,
523 matched mass flow and matched pumping power rank the fluids differently
524 in magnitude and, for pumping power at 18 bar, in sign. The effectiveness–
525 NTU pair inverts the engineering conclusion because it rewards a small ca-
526 pacity rate. A cryogenic coolant comparison must therefore state its basis
527 and its operating point, as Gačnik et al. found necessary for nitrogen-cooled
528 terminals [29]; the operating maps of Fig. 8 are offered as the form such a
529 comparison should take.

530 *Ideal and real ledgers disagree, and the disagreement is informative..* The
531 ideal ledger prefers nitrogen for a bare shield and helium for an insulated
532 one; the hardware ledger prefers helium in both cases, by 36–51 % at the
533 nominal shield temperatures and by 13–56 % at the coupled operating points
534 of Table 6, because the 4 K stage of a commercial cooler converts each milli-
535 watt of penalty into 5–9 W at the wall. The nitrogen route retains what Li et
536 al. identified three decades ago [11]: a constant shield temperature, stored re-
537 serve, and independence from the cryocooler, which now also means freedom
538 from its vibration [46]. Those are operational arguments; the thermodynamic
539 one is conditional on the harness.

540 *Methodological findings..* Two travel beyond this geometry. First, view-
541 factor matrices generated for low-emissivity enclosures must be checked for
542 closure and, if the enclosure is closed, row-normalised, which restores closure
543 though not reciprocity; the required closure accuracy is of order the emissiv-
544 ity, and agglomeration refinement does not supply it, whereas the normalised
545 load converges under it. The benchmark practice of Končar et al. [25] should
546 be extended by this test. Second, the mixing-cup temperature on a graded
547 mesh must be mass-flux weighted without an additional area factor; the cor-
548 rected accounting closed the energy balance of every run from a few per cent
549 to below 0.2 %.

550 *Limitations..* The enclosure imposes fixed temperatures on its plates and
551 shield and neglects contact resistance at the harness ends; a finitely conduc-
552 tive shield would carry the radiative-flux map of Fig. 7 into a temperature
553 map, which is the next step. The cold plate is a single-wall-heated smooth
554 duct with constant properties except for the helium density and the nitrogen
555 buoyancy term; conjugate conduction in the plate, which would spread the
556 wall temperature, is not resolved. Nitrogen is single phase by verified mar-
557 gin at 3 bar but not at 1 bar. The turbulent runs rely on the SST model at
558 $y^+ \leq 9.5$ and were not compared with experiment. The hardware efficiencies
559 are survey ranges from published data, and the split of a two-stage cooler's
560 efficiency between its stages is taken from those ranges rather than from
561 one machine; a measured capacity map of a specific machine would replace
562 them. Buoyancy in nitrogen was shown to be unsteady and was excluded
563 from the ranking rather than resolved; a transient calculation in a specified
564 plate orientation would quantify it.

565 7. Conclusions

- 566 1. A 77 K first shield costs essentially nothing at the first stage relative to
567 a 50 K shield: it intercepts 99.6 % of the radiation and 6.5 % less harness
568 conduction, 8.3 against 8.4 W for the enclosure studied. The penalty
569 appears at the 4 K stage, 106 mW, and is 86 % harness conduction
570 concentrated in the drive and flux bundles; it is worth 7.9 W of ideal
571 room-temperature power and 520–980 W with real 4 K coolers.
- 572 2. View-factor radiation in low-emissivity enclosures is controlled by ma-
573 trix closure: the generated matrix gave loads from 9.5 to 3.3 W under
574 agglomeration refinement against a 7.07 W closed form, and row nor-
575 malisation recovered the closed form to 4 % at the first stage and 0.5 %

- 576 at the 4 K stage, converging under refinement (7.36, 7.32 and 7.27 W
577 at 250, 400 and 800 faces).
- 578 3. As a first-shield coolant at 9.4 W, nitrogen at 3 bar gives 5 times he-
579 lium's heat-transfer coefficient in laminar flow and 6 times in turbulent
580 flow, and holds the plate within about 4 K of its inlet at $Re = 2300$
581 against helium's 21 K; helium at 1 bar lets the plate mean float above
582 77 K below $Re \approx 970$ and its hottest face above 77 K in every constant-
583 property laminar run (76.9 K at $Re = 2300$ with variable properties),
584 and needs 38 times the pumping power, but at 18 bar its pumping power
585 falls below nitrogen's by a factor of 8.5 with heat transfer unchanged at
586 fixed Reynolds number. At each fluid's optimum, the minimum pump-
587 ing power that holds the plate within its temperature limit, nitrogen
588 needs $1.0 \mu\text{W}$ at $Re = 500$; helium at 18 bar needs $0.6 \mu\text{W}$ if the plate
589 mean is the criterion and 4 to $31 \mu\text{W}$ if the hottest face is. The rank-
590 ing depends on the basis of comparison, the operating point and the
591 wall-temperature criterion.
- 592 4. The ideal ledger favours nitrogen by 21 % for a bare shield and helium
593 by 9 % for an MLI shield; with hardware efficiencies and the liquefac-
594 tion energy of the nitrogen consumed, the equivalent room-temperature
595 power favours helium by 36–51 % in both cases at the nominal shield
596 temperatures and by 13–56 % at the coupled operating points, an ad-
597 vantage that exists only where the helium plate is actually held near
598 50 K. At those efficiencies the nitrogen route becomes competitive when
599 the $77 \rightarrow 4$ K harness conduction is cut by about a factor of three, the
600 lever Eq. (8) identifies.
- 601 5. The comparison rests on code-verified numbers: NIST properties at a
602 stated reference state, a forced-convection basis with gravity-on sen-
603 sitivity, a mass-flux-weighted mixing-cup temperature that closes the
604 energy balance to 0.2 %, wall temperatures read from the wall itself,
605 and row-normalised view factors that reproduce the closed forms to
606 4 %. The conclusion that the shield choice and the harness design are
607 not separable follows from those numbers; experimental validation of
608 the cold-plate model is the next step.

609 CRediT authorship contribution statement

610 **Imerson Joao:** Conceptualization, Methodology, Software, Investiga-
611 tion, Formal analysis, Visualization, Writing – original draft, Writing – re-

612 view & editing.

613 **Declaration of competing interest**

614 The author declares no known competing financial interests or personal
615 relationships that could have influenced the work reported in this paper.

616 **Data availability**

617 All OpenFOAM cases (enclosure baseline and variants, cold-plate sweep,
618 grid studies), the post-processing and figure scripts, and the tabulated results
619 are available at
620 github.com/Imerson/cryostat-first-shield-He-vs-LN2; an archived copy will be de-
621 posited on Zenodo on acceptance.

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625 **Declaration of generative AI in scientific writing**

626 During the preparation of this work the author used Claude (Anthropic)
627 to assist with editing, revision and consistency checking of the manuscript
628 and with scripting the simulation campaign. The author reviewed and edited
629 all content and takes full responsibility for the published article.

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